



Module 1: License & Rules

FCC Framework · Licensing · Station Operation
2026–2030 Technician Question Pool



What We're Building Toward

- 5 modules $\times$ 2 hours = your FCC Technician license
- Module 1 (today): License & Rules — the FCC framework
- Module 2: Operating & Propagation — how to get on the air
- Module 3: Technical Core — the math and components
- Module 4: Signals, Antennas & Safety — the hardware
- Module 5: Capstone — practice exam + review $\rightarrow$ VE exam on-site
- 35 questions on the exam, 26 to pass (74%) — you WILL pass



The Exam — What to Expect

- 35 multiple-choice questions, randomly drawn from the question pool
- One question from each of 35 groups (T1A through T0C)
- 26 correct answers to pass = 74%
- No trick questions — the answer is always one of the four choices
- Most people finish in 15–20 minutes
- VE exam right here at the lab — bring photo ID and your FRN
- \$35 FCC fee is paid online through CORES after VEs submit your application



Let's Set Up HamStudy

Your #1 study tool between sessions



HamStudy Setup — Do This Now

- Go to hamstudy.org on your phone or computer
- Create a free account
- Select the 2026–2030 Technician question pool
- Scan the QR code on your handout (or visit the link) to share progress
- Share link: hamstudy.org/share?shareTo=qslujd
- This lets your instructor see which sections you're strong and weak on
- Between sessions: drill T1A–T1F to 80%+ confidence



Part 1: What Is Amateur Radio?

Subelement T1A · 1 exam question from this group



The Amateur Radio Service

- A radiocommunication service — not a product, not a toy, not CB
- Regulated by the FCC (Federal Communications Commission)
- Purposes defined in FCC Part 97:
 - Advancing skills in technical and communication phases of radio
 - Self-training and technical investigation
 - Emergency communications and public service
 - Building international goodwill



Key Terms You Need to Know

- Harmful interference — seriously degrades or repeatedly interrupts another service
- Beacon — a station transmitting for propagation observation (28.200–28.300 MHz on 10m)
- RACES — Radio Amateur Civil Emergency Service (requires FCC license AND civil defense certification)
- Frequency coordinator — recommended by local amateurs, coordinates repeater frequencies
- Willful or malicious interference — the single most explicitly prohibited act in amateur radio



The Phonetic Alphabet

- Alpha Bravo Charlie Delta Echo Foxtrot Golf
- Hotel India Juliet Kilo Lima Mike November
- Oscar Papa Quebec Romeo Sierra Tango Uniform
- Victor Whiskey X-ray Yankee Zulu
- Practice: spell your call sign (or your name) phonetically
- You'll use this **CONSTANTLY** on the air
- The FCC encourages (not requires) phonetics on phone



Part 2: Frequency Allocations

Subelement T1B · 1 exam question from this group



Technician Band Privileges

VHF / UHF (full privileges)

- 6 meters: 50.0–54.0 MHz
- 2 meters: 144.0–148.0 MHz
- 1.25 meters: 222.0–225.0 MHz
- 70 cm: 420.0–450.0 MHz
- Plus 902 MHz and up
- Max power: 1500W PEP

HF (limited privileges)

- 10 meters only: 28.000–28.500 MHz
- CW/digital: 28.000–28.300
- Phone/SSB: 28.300–28.500
- Max power on 10m: 200W PEP
- Can use FT8, CW, SSB, FM
- Can contact the ISS on VHF!



Important Frequency Rules

- Don't transmit on the exact edge of a band — sidebands extend beyond
- CW-only segments: 50.0–50.1 MHz and 144.0–144.1 MHz
- 2-meter FM simplex calling frequency: 146.520 MHz — memorize this
- Secondary allocations: you must not interfere with the primary user
- Technicians CAN use digital modes (FT8) on 10m, 6m, and 2m
- Any Technician can contact the International Space Station on VHF



Part 3: Licensing

Subelement T1C · 1 exam question from this group



Three License Classes

- Technician — your entry point. Full VHF/UHF + limited 10m HF
- General — adds most HF bands. Requires Element 2 + Element 3
- Amateur Extra — all frequencies. Requires Elements 2 + 3 + 4
- Each upgrade opens more bands — you keep all previous privileges
- No Morse code requirement for ANY license class
- Vanity call signs available to ANY licensed amateur (not just Extra)



License Numbers to Know

- License term: 10 years
- Renewal window: up to 90 days before expiration
- Grace period after expiration: 2 years — but you CANNOT transmit
- After passing: transmit as soon as your call appears in the FCC ULS database
- How you're notified: email from the FCC with a link to download your license
- If the FCC can't reach you by email: they can revoke your station license
- Keep your email address current in the FCC system!



Part 4: What You CAN and CAN'T Do

Subelement T1D · 1 exam question from this group



Transmissions: Allowed vs. Prohibited

PROHIBITED

- Broadcasting (transmitting to the general public)
- Obscene or indecent language
- Music (with narrow exception)
- Encryption to hide meaning (mostly)
- Business communications for profit
- Willful or malicious interference

ALLOWED

- Selling amateur equipment (occasionally, not regularly)
- One-way: beacons, telecommand, emergencies
- Compensation when part of classroom instruction
- International comms (unless country objects via ITU)
- Test transmissions (but you must still identify)
- Emergency communications override most rules



The Broadcasting Definition

- Broadcasting = transmissions intended for reception by the general public
- This is NOT what amateur radio does — we communicate station to station
- One-way transmissions that ARE allowed:
 - Beacons (propagation observation)
 - Telecommand of model craft (no ID required for this one!)
 - Emergency communications
 - Brief test transmissions (must still identify)
- When in doubt: if it's two-way between licensed operators, it's probably fine



Part 5: Control Operators

Subelement T1E · 1 exam question from this group



What Is a Control Operator?

- The licensed amateur designated by the station licensee
- Responsible for all transmissions and FCC compliance
- An amateur station may NEVER transmit without a control operator
- The station licensee designates who the control operator is
- Both the control operator AND the licensee share responsibility
- Transmitting privileges = the control operator's license class
- A Technician can NEVER be control op in an Extra-only segment



Control Types

- Local control — you're physically at the transmitter
 - → Example: sitting at your radio desk
- Remote control — you're somewhere else, controlling via a link
 - → Example: operating your station over the internet
- Automatic control — no operator present, station runs on its own
 - → Example: a repeater
- The control point = where the control operator function is performed
- Any station may be remotely controlled



Part 6: Station Identification

Subelement T1F · 1 exam question from this group



How to Identify Your Station

- Identify at least every 10 minutes AND at the end of the communication
- Use your FCC-assigned call sign
- On phone: identify in English (CW or phone emission)
- Tactical call signs are fine — but still give your FCC call every 10 min
- Self-assigned indicators: 'stroke,' 'slant,' or 'slash' — all acceptable
- You must ID even during test transmissions
- Exception: transmitting to control model craft (no ID required)



Repeaters, Third-Party & Club Stations

- Repeater = simultaneously retransmits on a different channel
- If a repeater retransmits a violation: the originating station is responsible
- Third-party communication = a message on behalf of another person
- You can let an unlicensed person talk — but YOU are responsible
- With foreign stations: only with third-party agreement countries
- Club station license requires at least 4 members
- FCC can inspect your station at ANY time — no advance notice



Quick Reference

The numbers that matter



Key Numbers for the Exam

Rules & Licensing

- Regulating agency: FCC
- License term: 10 years
- Grace period: 2 years (no TX!)
- Renewal window: 90 days before
- Station ID: every 10 min + end
- ID language on phone: English
- Club station: minimum 4 members

Frequencies & Power

- 2m calling freq: 146.520 MHz
- Tech max power (VHF+): 1500W PEP
- Tech max power (10m HF): 200W PEP
- CW-only: below 50.1, below 144.1
- Beacons on 10m: 28.200–28.300
- Control operator: ALWAYS required
- Vanity calls: ANY licensed amateur



Checkpoint Quiz

10 questions from the T1 question pool



Which of the following is part of the Basis and Purpose of the Amateur Radio Service?

- A. Providing personal radio communications for as many citizens as possible
- B. Providing communications for international contesting
- C. Advancing skills in the technical and communication phases of the radio art
- D. All these choices are correct



Which agency regulates and enforces the rules for the Amateur Radio Service?

- A. ARRL
- B. Homeland Security
- C. The FCC
- D. All these choices are correct



Which frequency ranges are available for phone operation by Technician licensees?

- A. 28.050 MHz to 28.150 MHz
- B. 28.100 MHz to 28.300 MHz
- C. 28.300 MHz to 28.500 MHz
- D. 28.500 MHz to 28.600 MHz



What is the normal term for an FCC-issued amateur radio license?

- A. Five years
- B. Eight years
- C. Ten years
- D. Life



Which of the following is prohibited?

- A. International communications on VHF
- B. Willful or malicious interference
- C. Third-party traffic using digital modes
- D. All these choices are correct



When may an amateur station transmit without a control operator?

- A. When using automatic control
- B. When the licensee is away and another amateur is using the station
- C. When the station is an auxiliary station
- D. Never



How often must you identify with your FCC-assigned call sign?

- A. At the beginning of each contact and every 10 minutes
- B. At least once during each transmission
- C. Every 15 minutes and at the end
- D. Every 10 minutes during and at the end of a communication



What determines the transmitting frequency privileges of an amateur station?

- A. The frequency coordinator
- B. The frequencies printed on the license
- C. The highest class on the premises
- D. The class of the control operator



What is the grace period for renewal if an amateur license expires?

- A. Two years
- B. Three years
- C. Five years
- D. Ten years



How does the FCC define broadcasting for the Amateur Radio Service?

- A. Two-way transmissions by amateur stations
- B. Any transmission made by the licensed station
- C. Transmissions directed only to amateur operators
- D. Transmissions intended for reception by the general public



Before Next Session

Your assignment for Module 2



Your Assignment

- Drill HamStudy sections T1A through T1F
- Target: 80%+ confidence on each section
- HamStudy tracks what you're getting wrong and drills those harder
- Trust the system — it works
- Coming in Module 2: Operating & Propagation
 - How to actually use a repeater
 - CTCSS tones, DMR, Q signals
 - How radio waves travel and propagation modes
 - Station setup and controls
- See you next session!